

Bernard Lappay

blappay1721@gmail.com | San Diego, CA | [blappay1721.github.io](https://github.com/blappay1721) | linkedin.com/in/blappay1721 | github.com/blappay1721

WORK EXPERIENCE

Silent Studios

Jun. 2026 – Present

Technical Contributor, Mineshoku Tensei

Remote

- Backend infrastructure, gameplay systems, and internal tooling for a live-service multiplayer RPG running across multiple server instances (Java, Python, Linux).
- Diagnosed a spatial-query bottleneck in a core gameplay system and built a Python computational-geometry simplification tool that cut the affected data's complexity by **~86%**.
- Designed real-time boss encounters: multi-stage attack sequences driven by timing-based state machines with ordered failsafes that prevent race conditions.
- Closed a time-of-check/time-of-use exploit in a core gameplay interaction, then rolled the fix across dozens of config files with a scripted, diff-verified patch.
- Diagnose JVM dependency and version-compatibility failures across many interoperating components; keep a ledger of confirmed platform regressions.

PROJECTS

Adaptive Math-Reasoning Inference Pipeline

CSE 151B Kaggle Competition, Spr. 2026

- Improved a fixed 4B reasoning model (Qwen3-4B-Thinking-2507) on math through inference-time design rather than retraining; ranked **#29** in a class-wide Kaggle competition.
- Built verifier-guided majority voting (3 candidates, 2-of-3 agreement, answers normalized across multiple-choice and free-response) with adaptive retries that escalate the token budget and re-prompt with prior attempts and verifier feedback.
- Audited and repaired 7,500 distilled training examples and trained LoRA SFT/DPO adapters; deployed the adapter only as a fallback for unresolved multiple-choice questions, where it helped.
 - **Key Results:** 68/100 on the final evaluation vs. 61/100 for voting and retries alone; voting and retries lifted multiple-choice accuracy from 73.6% to 81.6% over the base model.
 - **Stack:** Python; vLLM; Hugging Face; LoRA; checkpointed batch inference on RunPod H100s

Handwritten Digit Recognition

Python, NumPy, JavaScript

- Built a 1-NN classifier from scratch over 7,500 MNIST images reaching **95.4%** test accuracy; one vectorized distance computation classifies all 1,000 test digits in ~0.25 s, benchmarked against scikit-learn's KD-tree and ball tree.
- Shipped a browser demo that runs the same search in JavaScript: draw a digit and see its five nearest training images.

LFSR Message Encryptor

SystemVerilog, ModelSim

- Built a hardware stream cipher: a 6-bit LFSR with loadable taps and seed generates the keystream, and a program-counter-driven sequencer reads its configuration from a 256×8 data memory, writes a scrambled preamble, then encrypts the message.
- Verified all **64** output bytes against a software model of the same encryption in a self-checking testbench.

Self-Hosted Game Server Infrastructure

Personal Homelab

- Run a multi-server Minecraft network for **30+** players on a headless Ubuntu mini-PC: several Paper backends behind a Velocity proxy.
- Hardened it with ufw and fail2ban: rate-limited SSH, admin-only ports, and a custom jail whose filter I rewrote to catch slow read-timeout bot scanners; verified reachability end to end with ss, tcpdump, and external probes.
- Traced recurring NVMe timeouts, controller resets, and a watchdog crash to a firmware power-state (APST) bug and fixed it with a kernel boot parameter; traced mass player disconnects to Wi-Fi power management and mesh roaming.

EDUCATION

University of California, San Diego

Expected Jun. 2027

B.S., Computer Engineering

La Jolla, CA

- **Relevant Coursework:** Machine Learning: Learning Algorithms; Deep Learning (CSE 151B); Probabilistic Reasoning in AI; Computer Design and Architecture (Verilog, C, MIPS); Data Networks; Data Structures (Java)

TECHNICAL SKILLS

- **Languages:** Python, Java (4 yrs); C; JavaScript; SystemVerilog, Verilog; MIPS Assembly; MATLAB; Bash
- **ML & AI:** vLLM; Hugging Face; LoRA fine-tuning (SFT, DPO); RAG; Ollama; NumPy; scikit-learn; cloud GPUs (RunPod)
- **Linux & Networking:** Ubuntu Server; systemd; NetworkManager; rsync; tmux; PM2; SSH/SFTP; ufw; fail2ban; iptables; tcpdump
- **Hardware & Tools:** ModelSim; Quartus; self-checking testbenches; Git; Maven; Vim; VS Code; LaTeX